Scalability & Future Plan

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The project currently uses a fixed synthetic healthcare CSV dataset containing 5,000 records.	New patient information is not automatically updated, limiting real-time analysis.	High
2	The Tableau dashboard mainly provides descriptive analysis through charts, KPI cards, and filters.	It cannot currently predict future diseases or patient health risks.	Medium
3	The Flask website provides basic project information and dashboard access.	Advanced features such as user authentication and personalised dashboards are not available.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Flask website is designed for basic access and demonstration.	Deploy the application on a cloud platform with a production server to support more users.
2	Data Storage	Healthcare data is stored in a CSV file and analysed using Tableau Extract.	Integrate a database such as MySQL or PostgreSQL to manage larger healthcare datasets.
3	Performance	Tableau dashboards analyse the available dataset of 5,000 records.	Optimise data processing, use database connections, and improve dashboard loading performance for larger datasets.

4	Security	The current project is a basic demonstration website without advanced authentication features.	Introduce secure login, role-based access, encrypted communication, and appropriate data protection measures.

Future Roadmap:

Phase	Planned Feature / Enhancement Target Timeline Expected Impact
Phase 1	Complete the existing Tableau dashboard, Story, Flask website, testing, and documentation. Deliver the working Health360 project with interactive healthcare visualisations. Completed – October 2026
Phase 2	Integrate a database such as MySQL or PostgreSQL and support larger healthcare datasets. Improve data storage, management, and scalability.
Phase 3	Introduce machine learning models for patient risk analysis and predictive healthcare insights. Help identify potential health risk patterns using suitable validated data and models.
Phase 4	Deploy the Flask website on a cloud platform and introduce user authentication and advanced security features. Improve accessibility, security, and support for multiple users.